HEIRSTONE CONSULTING

HEIRSTONE CONSULTING RESEARCH | 2026

AGRO-INDUSTRIAL ZONES IN NORTH AND SUB-SAHARAN AFRICA

The crop comes first

Agro-industrial parks and agropoles: what they have delivered, and where processing in Africa actually happens



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SIX NUMBERS FROM THE REPORT

700

direct jobs at Ethiopia's Yirgalem park by late 2024, against more than 100,000 expected at its opening in 2021

55 on 55

companies on saleable hectares at Morocco's Souss Massa agropole, 90% commercialised

43%

of Côte d'Ivoire's record 2025 cashew crop processed at home, up from 21% in 2023

−58%

fall in Benin's cashew export earnings in 2024, the year its raw-nut export ban took effect

US$934m

committed by the AfDB to special agro-industrial processing zones: 27 sites in 11 countries (April 2025)

6%

of the 80,000 hectares allocated to the DRC's Bukanga-Lonzo park was planted at the 2014 peak, according to the Oakland Institute

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ABOUT THIS REPORT

This report examines agro-industrial zones, agropoles and special agro-industrial processing zones in Africa: how much land and money they have taken, how many tenants and jobs they have delivered, and how they compare with the places where crops are actually processed. It is written for agribusiness and food-processing boards, development finance institutions, governments and investment promotion agencies, zone developers and investors.

It draws on public sources current in 2026: zone operators, development-bank releases and appraisal reports, ICCO and UN Comtrade trade data, FAO, government statements and independent assessments. Where we combine published figures to produce a new one, such as financing per hectare or jobs per tonne processed, the calculation is labelled Heirstone analysis and its inputs are given.

WHAT WE FOUND

Processing has followed the crop and the policy. Most purpose-built zones are still waiting for both.

We built a register of 20 agro-industrial zones and programmes in 12 countries and set it against the trade record for cocoa, cashew, coffee, tea and oils. The zones that filled were small and phased, and sat close to a crop already moving in volume. The largest processing gains came from crop volume and policy.

1 **Job targets run far ahead of any processing workforce we can measure.**

Seven programmes in our register promise 22,000 to 400,000 jobs each. Côte d'Ivoire's cashew-processing industry, Africa's largest, employed about 15,000 people in 2024. Ethiopia's Yirgalem park was expected to create over 100,000 direct jobs when it opened in 2021; by late 2024 it had just over 700.

2 **Working agropoles sell about one hectare per firm.**

Morocco's Souss Massa agropole has 55 companies on 55 saleable hectares and is 90% sold. Ethiopia's four pilot parks cover 680 to 980 hectares and had 18 operating investors in 2024. A 15,000-tonne cashew plant in Benin occupies 1.7 hectares.

3 **One label covers two different businesses.**

Processing parks cost US$0.2m to US$1.1m per hectare; territorial agropoles spread US$10,000 to US$90,000 per hectare over farmland, irrigation and roads. The AfDB has committed over US$934m to 27 sites in 11 countries; outcomes by site are rarely published.

4 **Côte d'Ivoire built its processing industry largely outside agro-zones.**

It has out-ground the Netherlands in cocoa since 2020/21 and local cashew processing rose from 21% of the crop in 2023 to 43% in 2025 alongside export taxes, floor prices and nut allocation. Its pilot agropole hosts one cashew plant.

5 **Benin's export ban has yet to fill its factories, and much of the crop is reported to be crossing borders.**

Benin's raw-cashew export ban took effect in April 2024. Recorded cashew export earnings fell 58% that year; local processing reached about 50,000 t in 2025. The consultant Jim Fitzpatrick, quoted by Benin Web TV, estimates that nearly 70% of the 2024/25 crop went to Togo or Nigeria.

6 **At the one park with plant-level data, supply was the binding constraint.**

At Yirgalem in 2024 the avocado-oil plant ran at about 70% of capacity and another processor at 35% to 40%, and a dairy's milk intake had fallen to 300 litres a day from 2,000. Sub-Saharan Africa lost 23.0% of its food between harvest and retail in 2023, up from 22.1% in 2015 (FAO).

WHAT IT MEANS

Governments and agencies

Size zones to a proven crop, release land in 20 to 50 hectare tranches and publish tenants, tonnes and jobs every year.

Development finance

Release park money against tenants and tonnes processed, and appraise farmland programmes separately from processing parks.

Agribusiness and processors

Locate where the crop and the policy are. Secure raw material and working capital before floor space.

Zone developers and investors

Underwrite operating tenants per serviced hectare, and price the policy that sends the crop to the zone.

20

zones and programmes in the Heirstone Agro-Zone Register, across 12 countries

44

direct jobs per 1,000 tonnes of cashew processed in Côte d'Ivoire in 2024 (Heirstone analysis)

CHAPTER 1



01

Promises and payrolls

Agro-industrial parks are announced in hectares and jobs. This chapter judges them by operating tenants and people on the payroll.

When Ethiopia inaugurated the Yirgalem Integrated Agro-Industrial Park in March 2021, UNIDO reported that it was expected to create more than 100,000 direct jobs and 400,000 indirect ones. The park covers 294 hectares and represents over US$60m of investment. By late 2024, according to an IFPRI assessment, it had four operating investors, just over 700 direct jobs and 11 of a planned 152 sheds, covering 8.3 of 125 hectares of industrial area.

Yirgalem is one of four pilots. In the same assessment Bure had eight operating investors, Bulbula six and Baeker, in Tigray, none. Elsewhere on the continent the pattern of reporting is the same: approvals, budgets and job targets are published at launch, and later figures for tenants, occupancy and jobs are hard to find.

We went looking for the later figures. This chapter sets the promises against what we could measure. The gap runs to two orders of magnitude. Part of it comes from how targets are set, and part from how the parks have performed.

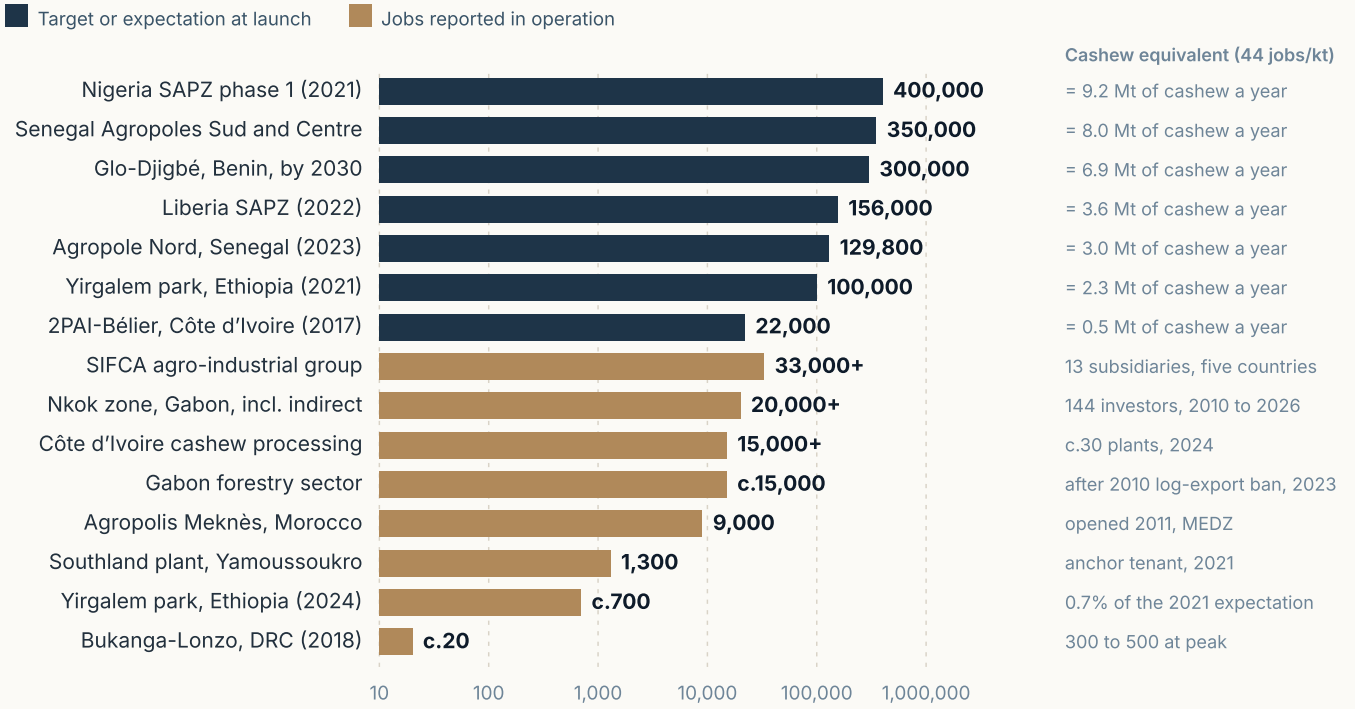
CALIBRATING THE TARGETS

Seven job targets, each larger than Côte d'Ivoire's entire cashew-processing workforce

EXHIBIT 1

Announced job targets are one to two orders of magnitude above the employment we can observe

Jobs targeted at launch or approval (navy) and jobs reported in operation (gold); logarithmic scale



Source: UNIDO (Mar 2021); IFPRI/CGIAR, Alemnew and Taffesse (Dec 2024); AfDB releases via TheCable (Dec 2021), AllAfrica (Nov 2023); AfDB Liberia SAPZ appraisal report (Sep 2022); Le Soleil (Oct 2025); GDIZ; AllAfrica FR (Feb 2017, Sep 2021); GSEZ; World Bank, Gabon Economic Update (Oct 2024); Fraternité Matin via allAfrica (Jun 2024); MEDZ; Groupe SIFCA; Oakland Institute (2019). **Heirstone analysis:** cashew equivalent = target ÷ 43.6 direct jobs per 1,000 t (15,000 jobs ÷ 344,026 t processed in Côte d'Ivoire, 2024). On 2023 tonnage the ratio is 56 per 1,000 t. Targets may include jobs outside the zone; Glo-Djigbé's covers all sectors, including textiles.

Job targets appear in almost every agro-zone announcement and are rarely tested against an observed benchmark. Our yardstick is Côte d'Ivoire's cashew industry: about 30 plants with 350,000 t of installed capacity employed more than 15,000 direct workers in mid-2024, 70% of them women, and processed 344,000 t of nuts that year. That is about 44 direct jobs per 1,000 t. Single plants bracket it: the Sino-Benin unit at Glo-Djigbé employs 500 people for 15,000 t a year (33 per 1,000 t) and Southland's plant at Yamoussoukro employed 1,300 for 24,000 t (54).

Used as a scale test, that intensity puts Nigeria's target of 400,000 direct jobs at the labour needed to shell about 9.2 Mt of cashew a year, more than twice the world crop of 4.2 Mt in 2024 (FAOSTAT). On 2023 tonnage the ratio is 56 per 1,000 t and the figure is still about 7 Mt. Yirgalem's 100,000 would need about 2.3 Mt, roughly one and a half times Côte d'Ivoire's record 2025 crop. Nigeria's hubs will handle many crops and its target covers jobs across hubs and transformation centres, so these figures show orders of magnitude only.

9.2 Mt

cashew a year that would employ 400,000 people at Ivorian labour intensity (about 7 Mt on 2023 tonnage); the 2024 world crop was 4.2 Mt

27x

Nigeria's phase 1 job target as a multiple of all direct jobs in Côte d'Ivoire's cashew plants (2024)

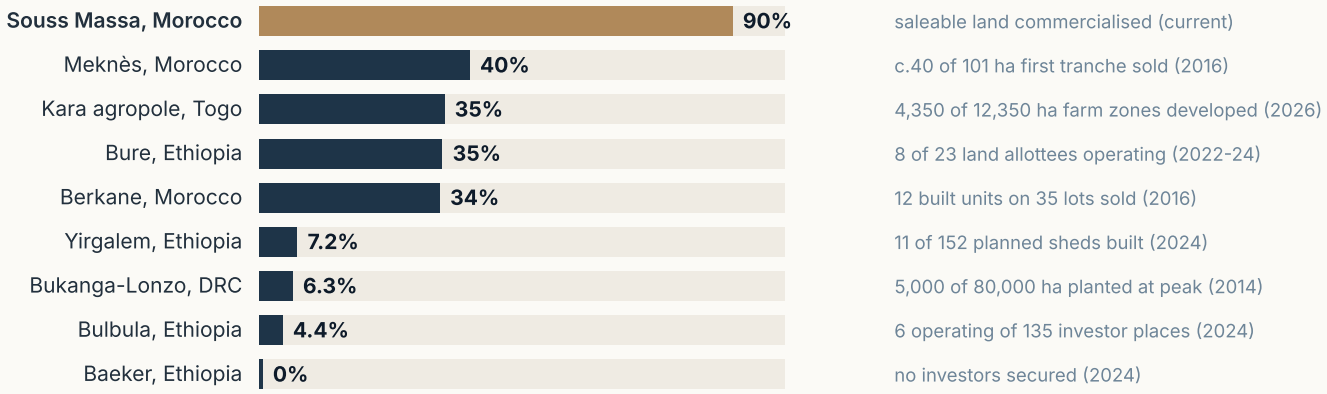
OCCUPANCY AND DENSITY

Only one zone we could measure is close to full

EXHIBIT 2

Ethiopia's pilot parks have filled under a tenth of their places; Souss Massa is 90% sold

Share of capacity sold, let, built or operating, using the measure each source publishes; latest date available



Source: MEDZ project pages (accessed 2026); Medias24 (Apr 2015); TelQuel (May 2016); AllAfrica FR (Aug 2026); The Reporter Ethiopia (Dec 2022); IFPRI/CGIAR (Dec 2024); Food Business MEA (Bulbula capacity, date not given); Oakland Institute (2019). **Heirstone analysis.** Measures differ by zone and are not strictly comparable.

Occupancy is reported in different ways, so Exhibit 2 uses whatever each source publishes. The spread is wide. Morocco's Souss Massa agropole reports 90% of its saleable land commercialised. Meknès had sold 35% to 40% of its first tranche four to five years after opening, and at Berkane only 12 of 35 sold lots carried a built unit in 2016. Ethiopia's parks sit at the bottom. Bulbula had six operating investors against room for 135, according to Food Business MEA, and Baeker had none.

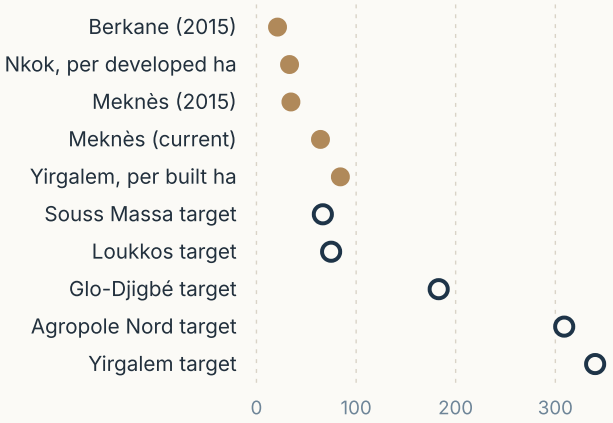
Cheap land has not filled them. Ethiopia's Investment Commission quotes land in its agro-industrial parks at US$1.2 per square metre a year, and Bure leased below US$1 in 2022. Meknès sold plots at MAD250 to 500 per square metre in 2016 (US$25 to 51 at that year's average rate) and now reports 9,000 jobs.

Exhibit 3 tests the targets against density. Moroccan agropoles report 21 to 64 jobs per hectare in operation and target 67 to 75. If all of Yirgalem's 700 jobs are set against its 8.3 built hectares, the sheds that exist are used at about 84 per hectare. The shortfall lies in the space that has not been built or let. Targets elsewhere assume 183 to 340 jobs on every hectare of the zone.

EXHIBIT 3

Targets assume three to five times the job density of Morocco's busiest agropole

Jobs per hectare: reported (gold) and targeted (ring)



Source: MEDZ; Medias24 (2015); GSEZ; IFPRI/CGIAR (2024); UNIDO (2021); GDIZ; AfDB via AllAfrica (Nov 2023). **Heirstone analysis:** jobs ÷ published hectares (Nkok includes indirect jobs; Agropole Nord uses 3 parks of 140 ha).

TIME AND MONEY

A naira budget set in 2021 was worth 27 cents on the dollar by 2024

The AfDB Board approved Nigeria's Special Agro-Industrial Processing Zones programme in December 2021 with US$210m of Bank financing. Ground was broken in Kaduna and Cross River in April 2025, 40 months later. Over that period the naira's official rate went from an average of NGN401 per US$ in 2021 to NGN1,479 in 2024, so a naira budget fixed at approval kept 27% of its dollar value.

States have raised naira budgets to keep pace: Kaduna's agriculture budget went from NGN1.4bn in 2023 to NGN74.2bn (US$47m at the AfDB's conversion) in 2025, alongside its SAPZ counterpart funding. The AfDB put phase 1 at US$510m in April 2025 (AfDB US$200m, IsDB US$150m, IFAD US$100m, GCF US$60m); Construct Africa reported US$538m with the federal share (2022).

Exhibit 4 shows the same effect across seven African currencies, where the loss over the same period ranged from under 10% to 73%. Zones financed in dollars but built with local counterpart money carry both the delay and the currency risk, so counterpart budgets need indexing or front-loading.

Exhibit 5 sets the money against the jobs. Appraisal figures imply US$125 to US$5,500 of financing per targeted job, a 44-fold spread driven mainly by the targets. A private cashew plant in Benin, Sino-Benin Noix de Cajou, cost about FCFA10bn (US$16.5m at the 2023 average rate) for 500 direct jobs, about US$33,000 each. Yirgalem's investment per direct job delivered so far is about US$86,000. Across the AfDB programme, US$934m of commitments over 27 sites averages US$35m of Bank money per site, or US$69m with co-financing.

40

months from AfDB Board approval (Dec 2021) to groundbreaking in Kaduna and Cross River (Apr 2025)

−73%

dollar value of a naira budget fixed in 2021, by 2024 (World Bank official rates)

x53

Kaduna State's agriculture budget in naira, 2023 to 2025

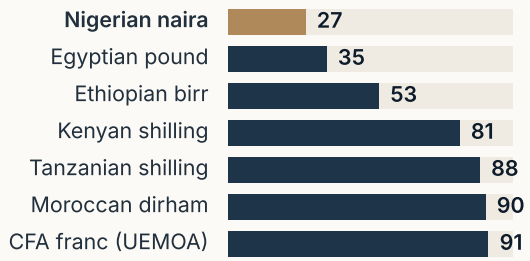
US$69m

average AfDB commitment plus co-financing per SAPZ site (27 sites, April 2025)

EXHIBIT 4

A local budget fixed in 2021 kept 27% to 91% of its dollar value by 2024

Dollar value in 2024 of 100 units of local-currency budget fixed in 2021; annual average official rates

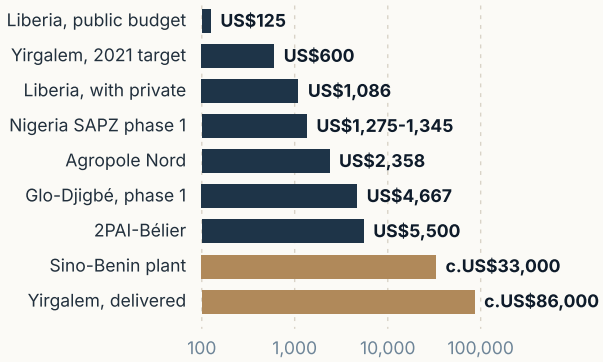


Source: World Bank WDI, official exchange rate PA.NUS.FCRF (accessed 2026). **Heirstone analysis:** 100 × rate 2021 ÷ rate 2024.

EXHIBIT 5

A working cashew plant cost about US$33,000 per job, far above appraisal ratios

Financing or investment per targeted job (navy) and per job delivered (gold), US$; logarithmic scale



Source: AfDB Liberia appraisal (Sep 2022); UNIDO (2021); IFPRI/CGIAR (2024); AfDB via AllAfrica (Apr 2025, Nov 2023); Construct Africa (Dec 2022); GDIZ; AllAfrica FR (Feb 2017); APA News. **Heirstone analysis:** financing ÷ job target; EUR at 2023 average (US$1 = EUR0.925); FCFA10bn at XOF606.6 per US$ (2023).

CHAPTER 2

02

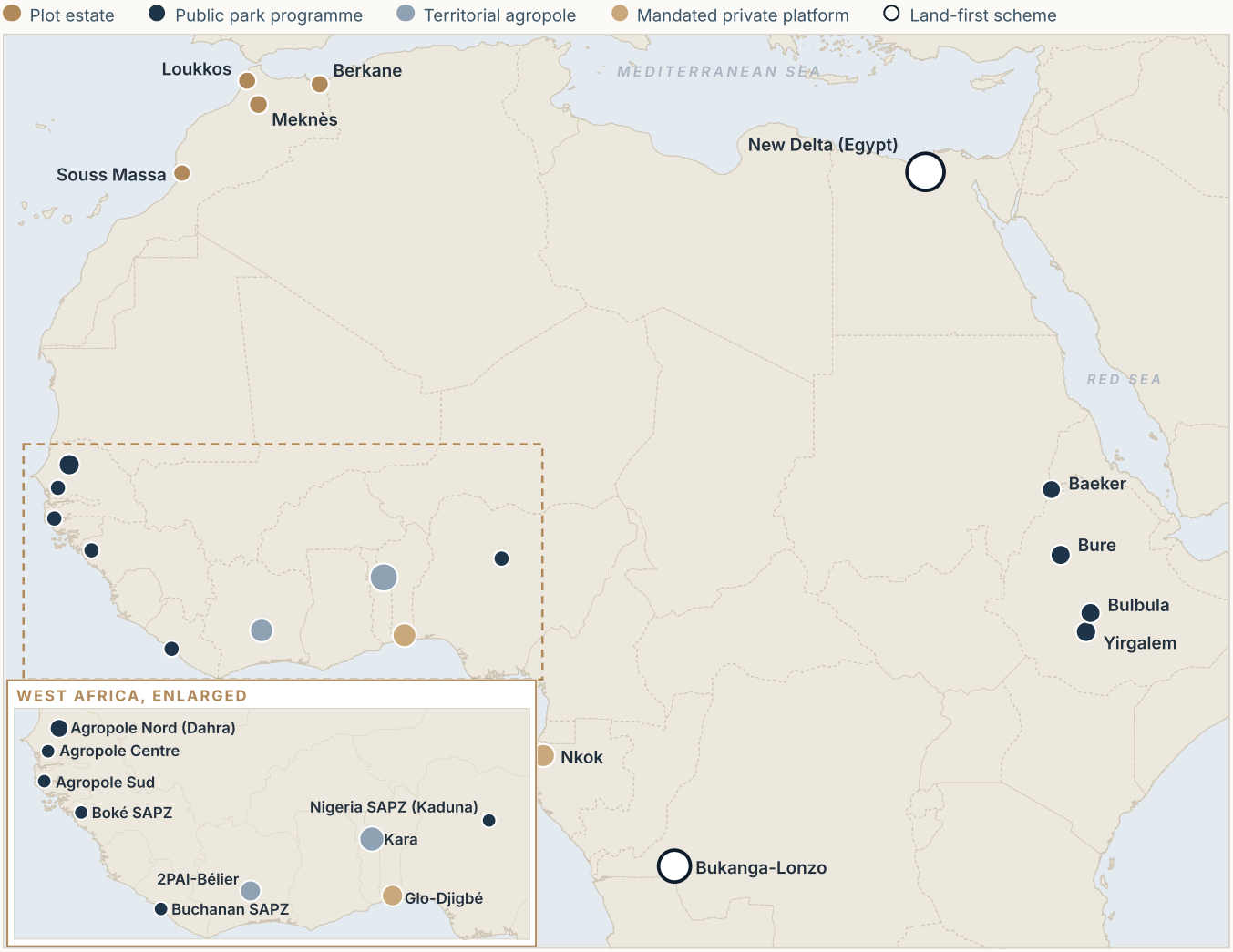
Twenty zones, five models

The Heirstone Agro-Zone Register brings together area, financing, tenants and jobs for 20 agro-industrial zones and programmes in 12 countries, from operator pages, development-bank releases, government statements and independent assessments. It is the base for the analysis that follows.

EXHIBIT 6

The register spans 12 countries, with clusters in Morocco, Ethiopia and francophone West Africa

Zones and programmes in the Heirstone Agro-Zone Register by model; marker size reflects published area (logarithmic); hollow markers mark land-first schemes



Source: Heirstone Agro-Zone Register (2026); sources in Exhibit 7. Locations are approximate town centres; programmes are shown at one representative town (Kaduna, Dahra, Ziguinchor, Kaolack, Boké, Buchanan). Bukanga-Lonzo and the New Delta are rough approximations of large areas. Base map: Natural Earth.

The register draws on four kinds of source: zone operators such as MEDZ, GSEZ and GDIZ; development-bank approvals and appraisal reports; government statements; and independent assessments such as IFPRI's 2024 review of Ethiopia's parks. Where they disagree on area or cost we show the range.

Its main gap is outcomes. Operators in Morocco and Gabon publish tenant counts. Most public programmes publish approvals and targets, and later figures come from researchers and the press. Nigeria's hub sites and areas, and tenant numbers at Glo-Djigbé, were not in the sources we could read.

THE HEIRSTONE AGRO-ZONE REGISTER

Eleven of the 20 zones are covered by a job target; seven report jobs, mostly through third parties

EXHIBIT 7

Targets and budgets are easy to find; tenants, tonnes and jobs are not

Area, financing, occupancy and jobs for 20 agro-industrial zones and programmes in 12 countries, latest public figures, 2026

ZONE, COUNTRY, YEAR OPENED OR APPROVED	MODEL	AREA (HA)	FINANCING OR INVESTMENT AS PUBLISHED	US$ PER HA	TENANTS AND OCCUPANCY	JOBS TARGETED	JOBS REPORTED
NORTH AFRICA							
Agropolis Meknès Morocco, 2011	Plot estate	140 (101 first tranche)	MAD300m+ (2015) to MAD500m+ (2016)	0.30-0.50m	34 projects on 35 ha sold (2015); 9 operating, c.30 installing (2016)	n/p	3,500 (2015); 9,000 (MEDZ)
Agropole de Berkane Morocco, 2012	Plot estate	102 (95.2 in 2016)	MAD220-250m (2015) or c.MAD360m (2016)	0.36-0.39m	35 lots sold, 12 units built (2016)	n/p	700 (2015); 600-1,000 in season
Agropole Souss Massa Morocco, 2017	Plot estate	75 (55 saleable)	n/p	n/p	55 firms installed or installing; 90% sold	5,000	n/p
Agropole du Loukkos Morocco	Plot estate	100 (50 first tranche)	n/p	n/p	35 clients installed or in progress	7,500	n/p
New Delta Egypt, 2026	Land-first scheme	c.924,000 (2.2m feddans)	EGP800bn (US$16.3bn)	17,600	n/p	10,000 direct	n/p
WEST AFRICA							
SAPZ phase 1 (7 states and FCT) Nigeria, 2021 / 2025	Public park programme	n/p	US$510-538m	n/p	Groundbreaking Apr 2025	400,000	n/p
Agropole Nord (Ogo, Bokhol, Dahra) Senegal, 2023	Public park programme	3 parks of 130-150	EUR283.05m (US$306m)	0.68-0.78m	Approved Nov 2023	129,800	n/p
Agropole Sud Senegal, 2025	Public park programme	n/p	XOF57bn; XOF109bn announced (US$187m)*	n/p	25 industrial sites planned, 2025-2029	350,000 with Centre	n/p
Agropole Centre Senegal	Public park programme	n/p	XOF108bn (US$186m)	n/p	In preparation	(with Sud)	n/p
2PAI-Bélier (Yamoussoukro zone) Côte d'Ivoire, 2017	Territorial agropole	39 zone; 1,356 farmland	XOF81bn (EUR123m; US$121m)	87,000	Anchor cashew plant; 300 business plans (2025)	22,000	1,300 at anchor (2021)
Kara agropole Togo, 2018	Territorial agropole	12,350 planned; 4,350 built	XOF77bn (US$125.8m)*	10,000-29,000	4 private operators; 7 processing centres (2026)	n/p	n/p
Glo-Djigbé Industrial Zone Benin, c.2021	Mandated platform	1,640	US$1.4-1.5bn phase 1	0.85-0.91m	5 cashew units of 100,000 t+ in all*	300,000 by 2030	n/p
SAPZ, Buchanan hub Liberia, 2021	Public park programme	n/p	US$19.45m	n/p	US$150m private investment targeted	156,000	n/p
SAPZ Boké and Kankan Guinea, 2026	Public park programme	n/p	US$30.52m (80% grant)	n/p	Ratified Mar 2026	n/p	n/p
CENTRAL AFRICA							
GSEZ Nkok Gabon, 2010	Mandated platform	1,126 (600 developed)	EUR740m invested	EUR0.66m	144 investors from 17 countries	n/p	20,000+ incl. indirect
Bukanga-Lonzo DRC, 2014	Land-first scheme	80,000 allocated; 5,000 planted	US$92-285m (sources differ)*	1,150-3,560	One operator, left 2017	n/p	300-500 peak; c.20 (2018)
EAST AFRICA							
Bure IAIP Ethiopia, 2021	Public park programme	155-264	US$169m	0.65-1.09m	23 allotted (2022); 8 operating (2024)	n/p	n/p
Yirgalem IAIP Ethiopia, 2021	Public park programme	109-294	US$60m	0.20-0.55m	4 operating; 11 of 152 sheds built (2024)	100,000	c.700 (2024)
Bulbula IAIP Ethiopia, c.2021	Public park programme	263-271	n/p	n/p	6 operating (2024); 135 places*	n/p	n/p
Baeker IAIP Ethiopia	Public park programme	151	n/p	n/p	No investors (2024)	n/p	0

Source: Operators, development banks, governments and press listed in the notes (chapter 2). **Heirstone analysis:** US$ per hectare = financing ÷ published area, at World Bank average rates for the year stated; ranges where sources differ. n/p: not published in the sources we found. *Single source: Agropole Sud budget (Ndarinfo); Kara cost (Milling MEA); Glo-Djigbé units (Benin Web TV); Bulbula places (Food Business MEA); Bukanga-Lonzo (Oakland Institute; Infos.cd). Senegal's 350,000 jobs cover Agropoles Sud and Centre together.

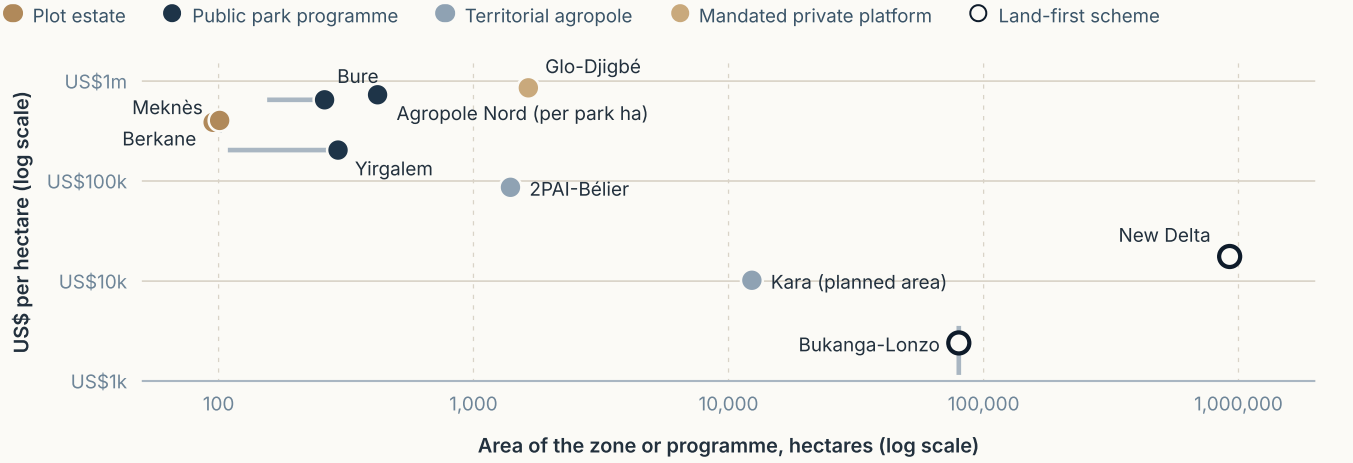
MONEY PER HECTARE

Parks cost US$0.2m to US$1.1m a hectare; territorial programmes, under US$90,000

EXHIBIT 8

Money per hectare separates the register into two families

Financing or investment per hectare against area, both on logarithmic scales; grey lines show ranges where sources give different areas or budgets



Source: Heirstone Agro-Zone Register (Exhibit 7). **Heirstone analysis.** Glo-Djigbé and Agropole Nord include private investment; Nkok (EUR0.66m per ha, cumulative) is not converted and not shown.

The development banks use one label, special agro-industrial processing zone, for both families, and in Exhibit 8 they sit at opposite corners. Parks of 100 to 1,640 hectares cost US$0.2m to US$1.1m per hectare. Programmes that spread money over farmland, rural roads and irrigation cost US$1,000 to US$90,000 per hectare of influence area.

The two need different appraisal. A park is judged on tenants and tonnes; a territorial programme on yields, farm incomes and the crop it puts within reach of a processor. Kara reports yield gains of 122% for maize, 75% for rice and 27% for soy. Exhibit 9 sets out the five models in the register.

EXHIBIT 9

Five models of agro-zone, and what each has delivered

Heirstone typology of the 20 zones in the register

MODEL	EXAMPLES IN THE REGISTER	SIZE	US$ PER HA	WHAT THE MONEY BUYS	RECORD SO FAR
Plot estate	Morocco's four MEDZ agropoles	75-140 ha, in tranches of 11-52 ha	US$0.30-0.50m	Serviced plots sold to processors	Souss Massa 90% sold; Meknès 9,000 jobs
Public park programme	Ethiopian IAIPs; SAPZs in Nigeria, Senegal, Liberia, Guinea	109-294 ha a park (Ethiopia); 130-150 ha (Senegal)	US$0.20-1.09m	Sheds, utilities, aggregation centres	0 to 8 operating investors a park (Ethiopia, 2024)
Territorial agropole	Kara (Togo); 2PAI-Bélier (Côte d'Ivoire)	1,395-12,350 ha, small industrial node	US$10,000-87,000	Irrigation, farm plots, rural roads	Kara yields up 27-122%; 4 private operators
Mandated private platform	Nkok (Gabon); Glo-Djigbé (Benin)	1,126-1,640 ha	US$0.85-0.91m (incl. tenants)	Private developer and power, with an export ban behind it	144 investors at Nkok
Land-first scheme	New Delta (Egypt); Bukanga-Lonzo (DRC)	80,000-924,000 ha	US$1,150-17,600	Land and irrigation first; processing later	New Delta inaugurated May 2026 with 8,100 centre pivots; Bukanga-Lonzo planted 6% of its allocation at peak

Source: Heirstone Agro-Zone Register; MEDZ; UNIDO; IFPRI/CGIAR (2024); AllAfrica FR (Aug 2026); GSEZ; GDIZ; Oakland Institute (2019); Egypt Today (2026). **Heirstone analysis.**

CHAPTER 3

03

Where the processing went

Africa's largest processing gains of recent years were in cocoa grinding and cashew shelling in Côte d'Ivoire, in plants that followed the crop.

In 2020/21 Côte d'Ivoire ground 620,000 t of cocoa beans and overtook the Netherlands (610,000 t) as the world's largest grinder, according to ICCO. It has stayed ahead: 793,000 t in 2022/23, 744,000 t in 2023/24 and a forecast 730,000 t in 2024/25, against 580,000 to 600,000 t in the Netherlands. In cashew, local processing rose from 40,383 t in 2016 to 659,579 t in 2025.

Neither gain depended on a flagship zone. In 2019 Reuters, citing the exporters' association GEPEX, put Ivorian grinding capacity at 712,000 t, with 12 active grinders. Cashew capacity sat in about 30 plants in 2024. The country's pilot agropole, 2PAI-Bélier, hosts one cashew plant on a 39-hectare industrial zone.

What the two crops share is volume. Côte d'Ivoire produced 1.7 Mt to 2.2 Mt of cocoa a year between 2022/23 and 2024/25 and marketed 1.55 Mt of cashew in 2025. In cashew, the processing share then rose alongside an explicit mix of export taxes, floor prices and nut allocation (Chapter 4). This chapter looks at what that processing earns, crop by crop.



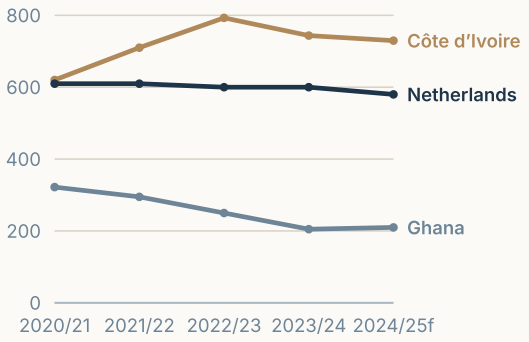
COCOA

Côte d'Ivoire grinds more cocoa than any other country and still earns twice as much from raw beans

EXHIBIT 10

Côte d'Ivoire has out-ground the Netherlands every season since 2020/21

Cocoa beans ground, thousand tonnes; 2024/25 forecast

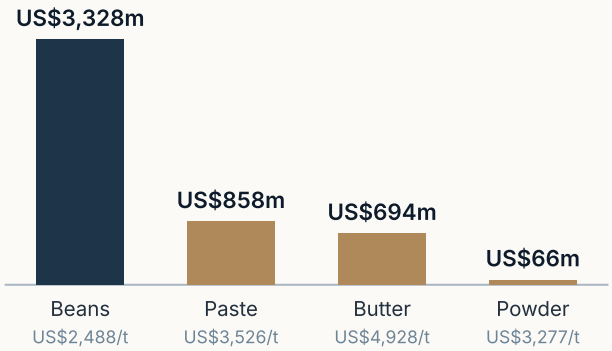


Source: ICCO, Quarterly Bulletin of Cocoa Statistics, Vol. XLIX No. 4 (Nov 2023) for 2020/21 and 2021/22; Vol. LI No. 1 (Feb 2025) for later seasons. 2022/23 revised from 770 to 793 (CIV) and from 280 to 250 (Ghana).

EXHIBIT 11

Processed cocoa earned a third of Côte d'Ivoire's cocoa export value in 2023

Export value by product, US$m, and unit value per tonne of product, 2023



Source: World Bank WITS (UN Comtrade), HS 180100, 180310, 180400, 180500, 2023. **Heirstone analysis:** unit value = value ÷ quantity; premium over beans per tonne of product, not per tonne of beans.

The 2023 trade data puts the scale of grinding in money terms. Côte d'Ivoire exported 243,219 t of cocoa paste, 140,932 t of butter and 20,287 t of powder, worth US$1,618m together, against 1,337,480 t of beans worth US$3,328m. Processed products were 33% of the value of these four lines. In 2024 bean exports alone reached US$3,994m as world prices rose.

Per tonne of product the premiums look large: 42% for paste, 98% for butter and 32% for powder over beans. Each tonne of beans yields less than a tonne of products, so the gain per tonne of beans processed is smaller than these figures suggest. We have not applied ICCO conversion factors.

The 'share of the crop processed' is a weak guide. Côte d'Ivoire's grind was 35% of its crop in 2022/23 and 44% in 2023/24, a jump that came from a 25% fall in the crop while grindings slipped 6%. Capacity use and tonnes ground are better measures. In 2017/18 Ivorian grinders used about 71% of their capacity, according to Reuters citing the exporters' association GEPEX.

Ghana shows the risk on the other side. Its grindings fell from 322,000 t in 2020/21 to a forecast 210,000 t in 2024/25. Africa produced about 72% of the world's cocoa in the 2024/25 forecast and ground 23% of the world's beans, or 31% of its own crop.

33%

share of processed products in Côte d'Ivoire's cocoa export value across beans, paste, butter and powder, 2023

35% to 44%

Ivorian grind as a share of the crop, 2022/23 to 2024/25; it rose when the crop fell

31%

of Africa's own cocoa ground in Africa, 2024/25 forecast (1,076 of 3,462 kt)

THE HEIRSTONE VALUE-CAPTURE TABLE

The premium for processing runs from 4% to fivefold, and the share processed rarely tracks it

EXHIBIT 12

Where processing pays per tonne, Africa still exports most of the crop in raw or bulk form

Export unit values of raw or bulk and processed forms, US$ per tonne of product, and processed share of export volume

COUNTRY	CROP	RAW OR BULK FORM	US$/T	PROCESSED FORM	US$/T	PREMIUM PER TONNE OF PRODUCT	PROCESSED SHARE OF EXPORT VOLUME	YEAR
Vietnam (benchmark)	Cashew	Raw nuts imported	1,159	Kernels exported	6,049	**+422%**	n/a	2023
Côte d'Ivoire	Cocoa	Beans	2,488	Butter	4,928	**+98%**	23% (all products)	2023
Ethiopia	Coffee	Green	5,189	Roasted	8,933	**+72%**	0.17%	2023
Tunisia	Olive oil	Bulk	3,992	Packaged	6,111	**+53%**	15.1%	2024/25
Côte d'Ivoire	Palm oil	Crude	878	Refined	1,256	**+43%**	50.9%	2023
Côte d'Ivoire	Cocoa	Beans	2,488	Paste (liquor)	3,526	**+42%**	23% (all products)	2023
Côte d'Ivoire	Cocoa	Beans	2,488	Powder	3,277	**+32%**	23% (all products)	2023
Kenya	Tea	Bulk (over 3 kg)	2,383	Retail packed	2,477	**+4%**	3.1%	2023
Egypt	Oranges	Fresh, US$795.7m		Juice, US$3.1m		n/a	0.4% of value	2023
Benin	Cotton	Lint, US$500.9m		T-shirts, US$0.4m		n/a	lint 1,240x by value	2024

Source: World Bank WITS (UN Comtrade), 2023 and 2024 exports by HS code; ONAGRI via Managers.tn (Oct 2025) for Tunisia, converted at TND3.107 per US$ (2024 average, World Bank). **Heirstone analysis:** unit value = value ÷ quantity. Premiums are per tonne of product: a tonne of kernels, butter or roasted coffee needs more than a tonne of raw material. Vietnam's premium is off the scale (▸).

Cashew carries the largest premium in the table. Vietnam's kernel exports averaged US$6,049 a tonne in 2023, 5.2 times the US$1,159 it paid for imported raw nuts. In aggregate the spread is thin. Vietnam imported 2.38 Mt of nuts worth US$2.76bn and exported 0.48 Mt of kernels worth US$2.92bn, about five tonnes in for each tonne out, before counting its own crop of 306,000 t (2024). The margin depends on how many tonnes of nuts each tonne of kernels takes, and on scale.

In coffee the value sits in the bean. Ethiopia exported 394 t of roasted coffee in 2023 against 234,881 t of green, and Uganda 50 kg. Ethiopia's green coffee earned US$5,189 a tonne that year, twice Uganda's US$2,578. In 2024/25 Ethiopia earned a record US$2.653bn on 469,000 t, 21% more than Uganda's US$2.2bn on 462,000 t. Grade and origin explain the gap; roasted coffee barely registers in either.

Packaging alone adds little. Kenya's retail-packed tea earned 4% more per tonne than bulk in 2023 and was 3.1% of export volume. Tunisia's packaged olive oil earned 53% more per kilogram than bulk in 2024/25 but was 15.1% of volume; Spain and Italy bought 53% of exports. Côte d'Ivoire already refines half its palm oil exports, at a 43% premium.

Two large exporters show the room for processing. Egypt, the world's largest orange exporter by volume, shipped US$795.7m of fresh oranges in 2023 and US$3.1m of orange juice. Benin exported US$500.9m of cotton lint in 2024 and US$0.4m of cotton T-shirts; Glo-Djigbé's textile park does not yet show in the data.

5.2x

Vietnam's kernel export price over its raw-nut import price per tonne, 2023

+6%

Vietnam's kernel export value over its raw-nut import bill, 2023 (US$2.92bn against US$2.76bn)

+4%

premium for Kenya's retail-packed tea over bulk, 2023

CHAPTER 4

04

Pushing the crop into the factory

Governments use three kinds of tool to keep a crop at home: taxes and floor prices that tilt the margin, allocation of raw material to local plants, and outright export bans. West and East Africa's cashew producers are using all of them, which makes cashew the clearest test of what each one does.

944,673 t

Côte d'Ivoire's 2024 cashew crop, the world's largest, ahead of India (794,910 t) and Tanzania (528,262 t), FAOSTAT

CÔTE D'IVOIRE

Two seasons took Côte d'Ivoire from 21% to 43%

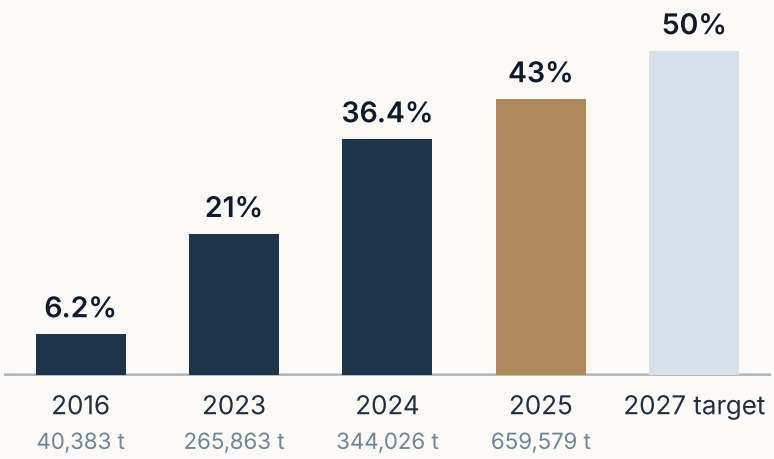
The Ivorian Cotton and Cashew Council reported that 6.22% of the crop was processed locally in 2016 and 21% in 2023. The share rose to 36.4% of a smaller crop in 2024 and to 43% of a record 1.55 Mt in 2025, when 659,579 t went to national processors. The target is 50% by 2027.

The policy mix has three parts. Raw nuts pay an export tax of US$0.15 per kg, according to the African Cashew Alliance. Floor prices are set along the chain: in 2025, FCFA425 per kg at the farm gate, 450 at interior warehouses, 479 at factory warehouses and 509 at the port, so the floor at a factory warehouse sits FCFA30 per kg (6%) below the floor at the port. Nuts are also allocated to local processors.

EXHIBIT 13

Local processing doubled in two seasons without stopping raw exports

Share of the cashew crop processed in Côte d'Ivoire and tonnes processed



Source: Conseil du Coton et de l'Anacarde via Connection Ivoirienne (Jun 2024), APA News (Jan 2025) and Fraternité Matin (Jan 2026). 2024 share is of marketed output; 2025 is 659,579 t of 1,549,221 t marketed. Team France Export puts 2025 at 600,000 t.

Raw exports continued throughout: more than 860,000 t of nuts left the country in 2025. After the record crop the 2026 farm-gate floor was cut 6% to FCFA400 per kg (US$0.69 at the 2025 average rate).

The industry has grown with the policy. In mid-2024 about 30 plants with 350,000 t of installed capacity employed more than 15,000 people, 70% of them women. West Africa as a whole converted 732,000 t of nuts into kernels in 2025, 51% more than a year earlier, and Côte d'Ivoire accounted for 82% of it, according to Team France Export. Of those plants, only Southland's at Yamoussoukro sits in an agro-zone in our register.

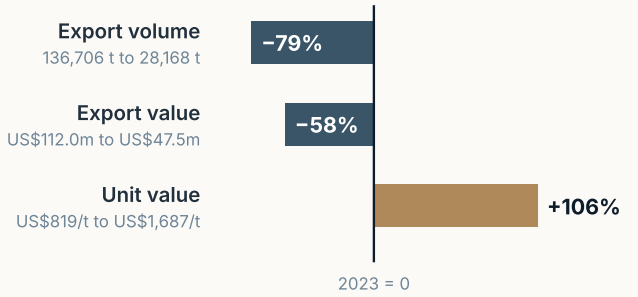
BENIN AND THE BAN

In the ban's first year, Benin's cashew exports fell by four-fifths in tonnes and 58% in value

EXHIBIT 14

Benin's cashew exports in 2024, the first year of the ban

Change on 2023, HS 080130 exports (raw nuts and kernels combined)

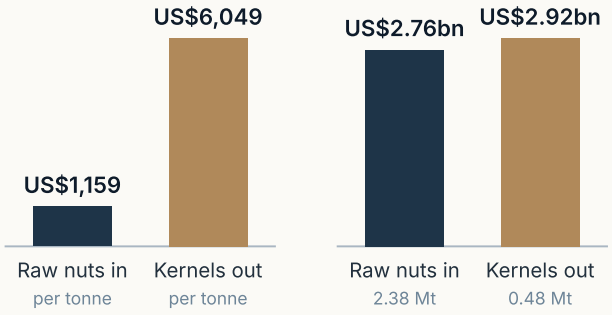


Source: World Bank WITS (UN Comtrade), Benin exports HS 080130, 2023 and 2024. **Heirstone analysis.** 2024 combines exports before and after the ban took effect on 1 April.

EXHIBIT 15

Vietnam earns five times more per tonne of kernels, and 6% more in total

Vietnam cashew trade, 2023: unit value (left) and total value (right)



Source: World Bank WITS (UN Comtrade), Vietnam imports and exports HS 080130, 2023. **Heirstone analysis.** Vietnam also grew 306,185 t of its own (FAOSTAT, 2024).

> *Benin had 130,800 t of cashew-processing capacity before the ban took effect and processed 17,700 t in 2023, on a consultant's figures.*

Benin chose a ban. Decree 2022-214 of 30 March 2022 prohibited raw cashew exports from 1 April 2024 and steered processing to the Glo-Djigbé Industrial Zone. Recorded cashew exports fell from 136,706 t worth US$112.0m in 2023 to 28,168 t worth US$47.5m in 2024, a 58% fall in earnings. The unit value doubled to US$1,687 a tonne, consistent with kernels replacing raw nuts in the mix.

Processing has grown. Benin converted about 50,000 t of nuts in 2025, against 13,000 t in 2020, according to Team France Export. Installed capacity was already 130,800 t in 2023, when 17,700 t were processed, according to the nut-market consultant Jim Fitzpatrick, quoted by Benin Web TV. Before the ban, plants were running at about 14% of capacity.

The same source estimates that less than 30% of the 2024/25 harvest was processed locally and nearly 70% was diverted to Togo or Nigeria. It puts Benin's farm-gate price at FCFA375 per kg against FCFA425 in Côte d'Ivoire, 12% lower, and reports five cashew units at Glo-Djigbé with at least 100,000 t a year of capacity between them, aiming for 200,000 t.

The contrast with Côte d'Ivoire is one of method. Both governments want nuts processed at home. The Ivorian route kept raw exports open and tilted prices toward the factory, and processed tonnes almost doubled in 2025. Benin's route banned raw exports, and on the consultant's estimate much of the crop still left the country.

Tanzania is the swing case. It produced a record 528,263 t in 2024/25, sold 406,362 t through its TMX auctions for TZS1.46 trillion (about US$562m at the 2024 average rate) and processed about 2% of its 2024 harvest at home. The planned Maranje park in Mtwara is to host plants with a cumulative capacity of 600,000 t, more than the record 2024/25 crop; a 2025/26 harvest of up to 700,000 t is forecast.

INSTRUMENTS COMPARED

Gabon's log ban built a timber industry, with a private zone and more than a decade

EXHIBIT 16

The instruments differ more in design than in ambition

Policy instruments used to push local agro-processing, with terms and outcomes as published by 2026

COUNTRY	CROP	INSTRUMENT	TERMS	OBSERVED OUTCOME
Côte d'Ivoire	Cashew	Export tax, floor prices, nut allocation, 50% target by 2027	US$0.15/kg tax; FCFA425/kg farm-gate floor (2025), 400 (2026)	Processing 21% (2023) to 43% (2025); raw exports above 860,000 t (2025)
Benin	Cashew	Raw-nut export ban, processing steered to Glo-Djigbé	Total ban from 1 Apr 2024 (decree of Mar 2022)	Export earnings −58% (2024); processing 13,000 t (2020) to 50,000 t (2025); c.70% diverted (consultant estimate)
Tanzania	Cashew	Export tax, minimum price, TMX auctions, planned park	US$0.15/kg tax; US$1.31/kg minimum price	About 2% processed (2024); Maranje park target 600,000 t
Burkina Faso	Cashew	Export tax and minimum price	US$0.90/kg tax; US$1.16/kg minimum	Not published
Mozambique	Cashew	Export tax	US$0.075/kg	Not published
Gabon	Timber	Log export ban and Nkok special economic zone	Total ban on unprocessed logs from 2010	Timber 3.2% of GDP and 6% of exports (2023); c.15,000 jobs; forest revenue x4 (2016-2023)
Ethiopia	Multi-crop	Public parks, rural transformation centres, tax incentives	Land US$1.2/m² a year; power US$0.045/kWh; tax exemptions	0 to 8 operating investors per pilot park (2024)
Nigeria	Multi-crop	Concessional DFI finance for processing hubs	US$510-538m, phase 1, 2022-2028	Groundbreaking Apr 2025; outcome data not yet published

Source: African Cashew Alliance, Cashew Barometer (2026, season not stated); Conseil du Coton et de l'Anacarde; Benin Web TV (May 2022, Jun 2026); World Bank WITS; Team France Export (Feb 2026); TanzaniaInvest; Ecofin (Sep 2025); World Bank, Gabon Economic Update (Oct 2024); Ethiopian Investment Commission; IFPRI/CGIAR (2024); Construct Africa (2022); AllAfrica (Apr 2025). Outcomes are not causal estimates.

Gabon is the case that ban advocates cite. It prohibited log exports and created the Nkok special economic zone in 2010. By 2023 the timber industry produced 3.2% of GDP and 6% of exports, forestry was the largest private employer with almost 15,000 jobs, and forest revenue to the state had risen fourfold since 2016, to CFA41.9bn (US$69m at the 2023 average rate), according to the World Bank.

Nkok, developed by Arise IIP, processed 1 million m³ of timber in 2022, exported EUR272m of products that year and shipped 23,297 containers (TEU) in 2023. It reports 144 investors from 17 countries on 600 developed hectares. The results took more than a decade and needed a private developer to host the new processors.

Benin's cashew policy follows the same design, a ban plus a zone. The early figures differ, and the crop may explain part of it: cashew is a smallholder crop sold in season, and the consultant's diversion estimate suggests that nuts crossed borders easily. Taxes, floor prices and allocation shift the margin without closing that route, and Côte d'Ivoire's raw exports continued while its processing doubled.

1m m³

timber processed at Nkok in 2022, with EUR272m of exports

x4

Gabon's forest revenue, 2016 to 2023 (World Bank)



CHAPTER 5

05

Feedstock before floor space

The agropoles that filled started small and sat next to a crop that already moved in volume. Parks built ahead of their supply struggle to keep processors running.

Plan Maroc Vert provided for a network of six agropoles: Meknès, the Oriental, Souss, Gharb, Haouz and Tadla. MEDZ, a CDG group developer, now markets four: Meknès, Berkane, Souss Massa and Loukkos. The early sites were slow. In 2015 Meknès had sold 35 of the 101 hectares in its first tranche, and in 2016 TelQuel reported nine units operating there and twelve built at Berkane.

The later sites were smaller from the start. Souss Massa opened 22 hectares in 2017 and a further 11 from July 2019; it now has 75 hectares developed, 55 saleable, 55 companies installed or installing and a 90% commercialisation rate. Loukkos released 50 of its 100 hectares first, offers plots from 2,000 m^2 with free-zone status, and has 35 clients. Its province grows 70,000 t of red berries and avocados.

Each was placed on a crop already sold in volume. The Moulouya plain behind Berkane grows close to 260,000 t of citrus on 20,000 hectares. Meknès, the earliest, had drawn about MAD1.3bn of private investment on 35 hectares sold by 2015, against a development cost of a little over MAD300m: about MAD4.3 of private money for every dirham spent on the site.

22 ha
Souss Massa's first phase, 2017

2,000 m^2
smallest plot offered at Loukkos

260 kt
citrus grown on the plain behind Berkane

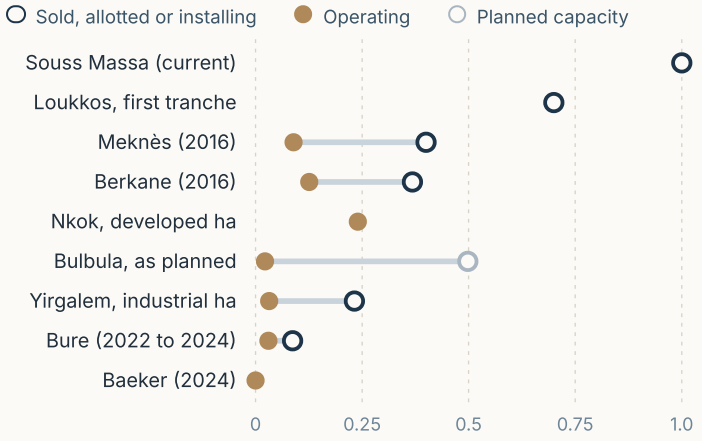
DENSITY

Working agropoles reach one firm per hectare; Ethiopia's parks were planned for half that and have under a tenth

EXHIBIT 17

Morocco's newer agropoles hold 0.7 to 1.0 firms per hectare; Ethiopia's pilots operate at 0.02 to 0.03

Firms per hectare: sold, allotted or installing (ring), operating (gold), planned capacity (grey ring)



Source: MEDZ (accessed 2026); TelQuel (May 2016); GSEZ; Food Business MEA (Bulbula capacity of 135 investors on 271 ha); IFPRI/CGIAR (Dec 2024); The Reporter Ethiopia (Dec 2022). **Heirstone analysis:** firms ÷ published hectares (Yirgalem: 125 ha of planned industrial area; Bure: 264 ha prepared).

1.7 ha

site of the 15,000 t a year Sino-Benin cashew plant at Glo-Djigbé, almost a tenth of Benin's crop

c.40 ha

needed for all 350,000 t of Côte d'Ivoire's cashew capacity at that footprint (Heirstone analysis)

c.980 ha

covered by Ethiopia's four pilot parks on inauguration and press figures

Agro-processors need less land than park plans assume. Souss Massa has one company for each saleable hectare and Loukkos 0.7 on its first tranche. The Sino-Benin cashew plant, which handles almost a tenth of Benin's crop, sits on 1.7 hectares. At that footprint the whole of Côte d'Ivoire's 350,000 t of cashew capacity in 2024 would fit on about 40 hectares.

Ethiopia's planning density was not unreasonable. Bulbula was designed, according to Food Business MEA, for 135 investors on 271 hectares, about half a firm per hectare. Yirgalem had supported 29 projects for 125 hectares of industrial area. The gap opened at the next step: by 2024 the four pilots had 18 operating investors between them on 680 to 980 hectares.

Phasing is what links the two. Souss Massa opened 22 hectares in 2017, added 11 in 2019 and is now 90% sold. Meknès and Berkane opened first tranches of about 100 and 52 hectares and took longer to sell them. A first phase of 20 to 50 hectares holds 20 to 50 firms at Moroccan density, a realistic launch cohort for most catchments.

Phasing also limits stranded cost. At US$0.2m to US$1.1m a hectare, every unlet hectare of park is a sunk public cost. A park built in three 50-hectare tranches can stop after the first if tenants do not come. A 300-hectare park built in one go carries the full cost from the start.

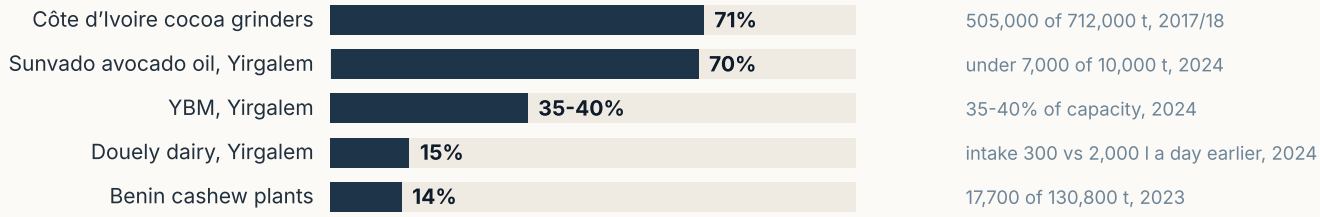
SUPPLY

Plants run below capacity when the crop does not reach them

EXHIBIT 18

Processors at Yirgalem ran at 35% to 70% of capacity; Benin's cashew plants at 14%

Capacity utilisation of selected processors and processing industries; for Douely, milk intake against earlier supply



Source: IFPRI/CGIAR, Assessment of Ethiopia's Integrated Agro-Industrial Parks (Dec 2024); Reuters via Business Recorder citing GEPEX (Feb 2019); Benin Web TV citing Jim Fitzpatrick (Jun 2026).

The IFPRI assessment of Yirgalem names the constraint. Sunvado's avocado oil plant processed just under 7,000 t against a potential 10,000 t. YBM ran at 35% to 40% of capacity, and the Douely dairy was receiving 300 litres of milk a day against 2,000 earlier; avocado and milk supply were short. These tenants did arrive, and their plants still stand idle for part of the year for lack of raw material.

Ethiopia's answer is the rural transformation centre, an aggregation point that buys and grades the crop for the park. One centre now serves each operating park, at Motta, Shashemene and Bensa, within a catchment UNIDO put at 100 km; the programme envisages 23 centres in 17 growth corridors. Benin's figures show the same constraint in a different form: 130,800 t of capacity in 2023, 17,700 t processed.

Losses make the catchment smaller than the harvest. FAO estimates that Sub-Saharan Africa lost 23.0% of food between harvest and retail in 2023, up from 22.1% in 2015 and the highest of any region; the Malabo Declaration had committed governments to halving post-harvest losses by 2025. APHLIS puts Africa's cereal losses at about 18.5 Mt a year, 9.7 Mt of it maize. UNEP and FAO estimate that 12% of the world's food was lost for lack of refrigeration in 2017.

23.0%

of food lost between harvest and retail in Sub-Saharan Africa, 2023 (FAO)

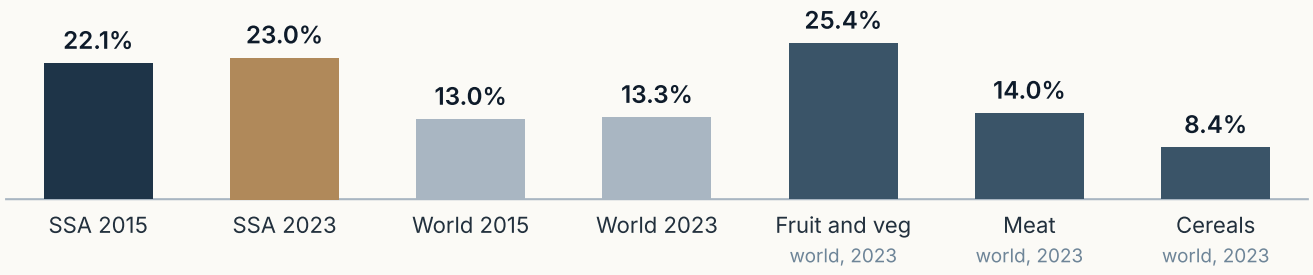
18.5 Mt

cereals lost after harvest in Africa each year, 2017 to 2021 average (APHLIS)

EXHIBIT 19

Sub-Saharan Africa's food loss is rising, and fruit and vegetables lose the most

Food lost between harvest and retail as a share of production, Sub-Saharan Africa and world, and world by product group



Source: FAO, SDG indicator 12.3.1a Food Loss Index data portal (accessed 2026).

CHAPTER 6

06

North Africa and Sub-Saharan Africa

Each country starts from a different crop, currency and zone model. The table sets out where each stands; the next three pages draw out what it means for North Africa, francophone West and Central Africa, and the anglophone West and East.

EXHIBIT 20

Eighteen countries, each starting from a different crop and zone model

Agro-zone model, crop position and the indicator we would track, by country

COUNTRY	ZONE MODEL IN PLACE	WHERE PROCESSING IS MOVING, OR COULD	NUMBER TO WATCH
NORTH AFRICA			
Morocco	Four MEDZ agropoles of 75-140 ha; six planned under Plan Maroc Vert	Fresh tomatoes 658,746 t exported (US$1.14bn, 2023); preserved olives US$136m	Status of the Gharb, Haouz and Tadla agropoles
Egypt	Land-first: New Delta, c.924,000 ha (EGP800bn); Mostaqbal Misr industrial zone*	Citrus c.2 Mt exported; record US$11.5bn of agri-food exports (2025)	Tenants and tonnes in agro-industrial zones
Tunisia	No agro-zone in the register	Olive oil 268,600 t exported in 2024/25, 15.1% packaged	Packaged share of olive oil exports
FRANCOPHONE WEST AND CENTRAL AFRICA			
Côte d'Ivoire	2PAI-Bélier, pilot of nine planned agropoles	Cocoa grind 730-793 kt; cashew 43% processed (2025)	Cashew share against the 50% target for 2027
Senegal	Four funded agropoles, XOF410bn (US$705m) at first budgets	North: 80% of national rice, 90% of industrial tomato	Tenants at Ogo, Bokhol and Dahra
Benin	Glo-Djigbé (1,640 ha) under the raw-cashew ban	Cashew 50,000 t processed (2025); lint 259,467 t exported (2024)	Cashew earnings and cross-border flows
Togo	Kara agropole (12,350 ha planned); Adétikopé platform (Arise IIP)	Rice, maize, soy; yields up 27-122%	Private operators beyond the first four
Guinea	Boké and Kankan SAPZ, US$30.52m (2026)	Not published	First tenants
Gabon	Nkok (1,126 ha, 144 investors)	Timber 1m m^3 processed (2022)	Share of developed land in use
DRC	Bukanga-Lonzo; 22 parks of 1.59m ha planned (2014)	Maize, peak 5,000 ha planted	Outcome of the 2022 revival
Cameroon	No agro-zone in the register	Cocoa 320 kt produced (2024/25 forecast)	Local grinding
ANGLOPHONE WEST AND EAST AFRICA			
Nigeria	SAPZ phase 1: 7 states and FCT; phase 2 for 28 states	Cocoa 350 kt; 49,507 t of raw cashew to Vietnam in May 2026	Tenants at hubs after the 2025 groundbreaking
Ghana	No agro-zone in the register	Cocoa grind down from 322 kt to 210 kt (2020/21 to 2024/25f)	Grinding recovery
Liberia	SAPZ, Buchanan hub, US$19.45m	Nine value chains named	Private investment against US$150m target
Ethiopia	Four pilot IAIPs; 23 RTCs; 17 growth corridors	Coffee US$2.653bn (2024/25); roasted 0.17% of volume (2023)	Operating investors per park
Kenya	County aggregation and industrial parks, 25 counties*	Bulk tea 545,574 t; cut flowers US$661m (2023)	Parks completed and let
Tanzania	Maranje park planned (600 kt capacity)	Cashew 528 kt (2024/25), c.2% processed	Processing share of the crop
Uganda	No agro-zone in the register	Coffee US$2.2bn (2024/25); roasted 50 kg (2023)	Grade premiums

Source: Heirstone Agro-Zone Register; ICCO (2025); World Bank WITS; Conseil du Coton et de l'Anacarde; ONAGRI; Ecofin (Jan 2026); Business Today Egypt (Jan 2026); AllAfrica (2023, 2025); African Cashew Alliance (Jul 2026); Ethiopian Investment Commission; IFPRI/CGIAR; Food Business MEA; Ecofin (Jul 2025, Sep 2025); TanzaniaInvest. *Press sources only.

NORTH AFRICA

Egypt runs the register's largest programme, Morocco sells plots and Tunisia ships olive oil in bulk

Morocco

Morocco has the only network in our register that publishes commercialisation rates, and the only one where a site is close to full. Its lesson is about size and sequence: Souss Massa opened with 22 hectares and added 11 two years later, and Loukkos sells plots from 2,000 m^2. The regional fund at Souss Massa adds an investment premium of up to MAD250 per m^2 (US$27 at the 2025 average rate). Morocco exported 658,746 t of fresh tomatoes worth US$1.14bn in 2023 and US$136m of preserved olives, so the export horticulture the agropoles serve is already large. The open question is the rest of the network: we could not verify the status of the Gharb, Haouz or Tadla agropoles planned under Plan Maroc Vert.

Egypt

Egypt's model is land first, and its New Delta is the largest programme in our register by area and by budget. Inaugurated on 17 May 2026, it covers about 2.2 million feddans (about 924,000 ha) at a cost of nearly EGP800bn, about US$16.3bn at the 2025 average rate. It is expected to create about 10,000 direct jobs and over 360,000 indirect ones, and is irrigated by 28 main pumping stations, 9,100 km of pipelines and 8,100 centre pivots fed by about 7.5 million m^3 a day of treated drainage water. The first phase of the Mostaqbal Misr industrial zone was inaugurated in 2025, according to Enterprise, and Milling MEA reports plans for a US$7bn agro-industrial complex with Chinese investors on 420,000 ha in New Valley.

The processing opportunity is already in the export basket. Egypt shipped a record US$11.5bn of agricultural and food products in 2025, up from US$10.7bn in 2024, with oranges, potatoes and sweet potatoes close to 40% of sales, and around 2 Mt of citrus. It was the world's largest orange exporter for a sixth consecutive year, according to FreshPlaza. Juice is the next step: orange juice exports were worth US$3.1m in 2023 against US$795.7m of fresh oranges. A small serviced plot estate near citrus packing and export routes would test that demand at low cost. Wheat is the other pull: consumption of about 21 Mt in 2023/24 against domestic output of about 9.1 Mt, according to Milling MEA, and the New Valley complex is planned for wheat, maize, soy and sunflower.

Tunisia

Tunisia exported 268,600 t of olive oil in the 11 months to September 2025, worth TND3,599m (about US$1.16bn at the 2024 average rate). Packaged oil was 15.1% of volume and 21.4% of value, and earned about TND19.0 per kg against TND12.4 for bulk. Spain and Italy bought 53% of exports. Doubling the packaged share to 30% at constant prices would add about TND260m (US$85m), 7% of export earnings (Heirstone analysis). Capturing it would take brands and distribution in the importing markets, led by Spain, Italy and the United States.

MAD4.3

private investment per dirham of development cost at Meknès by 2015 (MAD1.3bn against MAD300m)

360,000+

indirect jobs expected from Egypt's New Delta, alongside about 10,000 direct jobs

c.2 Mt

of citrus exported by Egypt in 2025, within a record US$11.5bn of agri-food exports

+53%

premium per kg for Tunisia's packaged olive oil over bulk, 2024/25

8,100

centre pivots and 9,100 km of pipelines irrigating the New Delta, fed by about 7.5 million m^3 a day of treated drainage water

FRANCOPHONE WEST AND CENTRAL AFRICA

Senegal has budgeted US$705m for four agropoles, and Benin has tied its cashew crop to one zone

Senegal

Senegal has funded four agropoles at XOF410bn on their first budgets (US$705m at the 2025 average rate): Centre XOF108bn, Sud XOF57bn, Nord XOF189bn and Ouest XOF56bn, with the Est unfunded. Agropole Nord, approved by the AfDB in November 2023, costs EUR283.05m and draws 27% of its financing from private sponsors as debt and equity. It plans three parks of 130 to 150 hectares at Ogo, Bokhol and Dahra and 14 aggregation and service centres, in a zone that grows over 80% of national rice and 90% of industrial tomatoes. Its target of 129,800 direct jobs by year five equals about 310 per hectare of park; at the 21 to 75 jobs per hectare reported or targeted in Morocco, the parks themselves would hold about 9,000 to 31,500. Ndarinfo reports a presidential announcement of XOF109bn (US$187m) for Agropole Sud, up from XOF57bn; the agropole plans 25 industrial sites in 2025-2029.

Côte d'Ivoire

2PAI-Bélier, the pilot of nine planned agropoles, is a territorial programme with a small industrial node: a 39-hectare zone at Yamoussoukro, 1,241 hectares of rice perimeters, 115 hectares of market gardens and 542 km of rural roads, for about XOF81bn. It was 85.7% complete in the latest report. By May 2025 it had validated 300 business plans worth XOF300.8m in total, about XOF1m (US$1,700) each, which makes it an incubator for small firms as much as a processing estate.

Benin and Togo

Glo-Djigbé covers 1,640 hectares, with US$1.4bn to US$1.5bn of phase 1 investment, a planned 143 MW power plant and a one-stop shop of 13 agencies. The ban decree steers cashew processing to the zone. Togo's Kara agropole, 423 km north of Lomé, has developed 4,350 of 12,350 planned hectares of farm zones, completed seven processing centres and installed four private operators; the 2018 plan aimed to lift the local processing share from 19% to 40%.

Guinea, Gabon, DRC and Cameroon

Guinea ratified US$30.52m of AfDB financing, 80% grant, for Boké and Kankan in March 2026. Gabon's Nkok shows a mandated platform at maturity. In the DRC, Bukanga-Lonzo was allocated 80,000 hectares in 2014 as the pilot of 22 parks covering 1,588,500 hectares; according to a 2019 Oakland Institute report, maize plantings fell from 5,000 hectares in 2014 to 2,000 in 2016. Reported public spending ranges from US$92m, the allocation cited in that report, to US$285m of Treasury disbursements reported by Infos.cd in 2022. A revival was announced in 2022. Cameroon, with 320,000 t of cocoa, has no zone in our register.

US$705m

four funded Senegalese agropoles at first budgets (XOF410bn)

27%

of Agropole Nord's EUR283m financing from private sponsors (19.3% debt, 7.7% equity)

c.310

direct jobs targeted per hectare of park at Agropole Nord, against 21 to 64 reported in Moroccan agropoles

35%

of Kara's planned farm zones developed by August 2026

US$102.79m

Green Climate Fund resources for SAPZ irrigation on 39,179 ha in Guinea, Senegal and Togo (2024)

ANGLOPHONE WEST AND EAST AFRICA

In Nigeria and East Africa, the parks have yet to reach the crops that are already exported

Nigeria, Ghana and Liberia

Nigeria's phase 1 covers seven states and the federal capital territory, with phase 2 planned for the remaining 28 states. Its hub sites, areas and anchor tenants are not yet published in the sources we could read. The crops are there: Nigeria produced about 350,000 t of cocoa in 2024/25 and shipped 49,507 t of raw cashew to Vietnam in May 2026 alone, worth N110bn (US$79m). Ghana's cocoa grind has fallen from 322,000 t to a forecast 210,000 t. Liberia's SAPZ costs US$19.45m, targets US$150m of private investment and 156,000 jobs, and was appraised at a financial return of 25.83%.

Ethiopia

Ethiopia offers low occupancy costs: land at US$1.2 per m^2 a year, sheds at US$1.5 per m^2 a month for foreign investors, power at US$0.045 per kWh and tax exemptions. Its parks are still short of tenants and crop. The country's largest export is moving on other terms. Coffee earned a record US$2.653bn in 2024/25, the year the minimum capital for exporters was raised from ETB1.5m to ETB15m. The birr went from ETB43.7 per US$ in 2021 to ETB82.6 in 2024. Two fertiliser projects are far larger than the home market: OCP's Dire Dawa plant (US$2.4bn, 2.5 Mt a year, rising to 3.8 Mt) has shown no progress since 2021 according to TelQuel, and Dangote announced a US$2.5bn, 3 Mt plant at Gode in 2025, against national use of about 700,000 t a year.

Kenya, Tanzania and Uganda

Kenya tendered county aggregation and industrial parks in 25 counties in 2023, and press reports cite KSh3bn of national support (US$23m at the 2025 average rate) and a KSh500m (US$3.9m) bill per park shared with counties. Its exports show where value already sits: 545,574 t of bulk tea worth US$1.30bn and US$661m of cut flowers in 2023, a cold-chain business more than a processing one. Tanzania is the clearest untapped case: 528,263 t of cashew in 2024/25, about 2% processed, and earnings of TSh1.52 trillion (about US$584m, Bank of Tanzania). At the Sino-Benin plant's cost of about EUR1,000 per tonne of capacity, processing the 518,000 t that leave raw would need about EUR520m of plant (Heirstone analysis). An AfDB loan approved in November 2025, US$24.6m to MeTL for sisal, macadamia and tea, went to a company; no zone is named. Uganda earned US$2.2bn from coffee in 2024/25, with Europe taking 69% of exports.

25.83%

financial rate of return in the AfDB's appraisal of Liberia's SAPZ (2022)

US$0.045

per kWh for power in Ethiopia's agro-industrial parks

3.6-5.4x

Ethiopia's fertiliser use covered by the OCP and Dangote plant capacities announced

EUR520m

plant needed to process Tanzania's raw-cashew surplus at Benin's cost per tonne (Heirstone analysis)

CHAPTER 7

07 The Heirstone Agro-Zone Test

Six questions to answer before the first shed is built. Each comes from a pattern in the register or the trade data and carries a threshold that can be checked at appraisal.

EXHIBIT 21

Six gates between a hectare of farmland and a working processing zone

The question, the threshold we would apply (bold) and the evidence behind it (grey)

1 CROP	2 POLICY	3 SIZE	4 ANCHOR	5 SUPPLY CHAIN	6 TARGETS
Is a marketed surplus within reach, large relative to the plants planned?	Does policy make local processing pay, predictably?	Is the first phase sized to the first cohort of tenants?	Is a processor or private developer committed before public money is spent?	Can tenants buy, store, finance and power their way through the season?	Are jobs and currency risk set at believable levels?
Planned intake no more than a tenth of the marketed crop in the catchment; zone capacity never above the crop	**A price edge for the factory set before the season, with raw exports still possible**	**First tranche of 20 to 50 ha; the next released at 70 to 80% sold or let**	**At least one signed anchor whose intake is matched to the crop**	**Aggregation, storage and power in the same budget; 60%+ utilisation by year three**	**No more than 75 jobs per ha or 56 per 1,000 t processed; counterpart funds front-loaded**
A 15,000 t cashew plant took almost a tenth of Benin's crop. Maranje targets 600 kt against a record 528 kt crop.	Côte d'Ivoire's factory floor FCFA30/kg below the port: 21% to 43%. Benin, first ban year: export earnings −58%.	Souss Massa: 22 ha, then 11 ha, now 90% sold. Ethiopia: 109-294 ha parks, 0 to 8 operating tenants.	Southland employed 1,300 at Yamoussoukro; Arise IIP at Nkok and Glo-Djigbé. Baeker: no investors.	Yirgalem plants at 35-70%. Benin: 17,700 t on 130,800 t of capacity. SSA food loss 23%.	Targets of 183-340 jobs per ha. Naira budget worth 27% in dollars after 40 months.

Source: Heirstone analysis, drawing on Exhibits 1 to 19. Thresholds are Heirstone guidance derived from the cases cited, not regulatory standards.

The test is sequential. A zone that fails the first gate cannot be rescued by the later ones: Bukanga-Lonzo had land, money and an operator, and at its 2014 peak planted 6% of its allocation. Where the first two gates are met, processing can grow with little help from a zone: Côte d'Ivoire's cashew plants located themselves where the crop and the price edge were.

Gates three to five decide cost and speed. Moroccan agropoles show what small, phased estates achieve, and the IFPRI work on Ethiopia shows that a tenant without a steady crop runs at a fraction of capacity. The sixth gate protects the credibility of the programme: a target set at three to five times observed density is likely to be missed even if the zone performs well.

The thresholds are deliberately simple so that a board, a lender or a ministry can apply them from the appraisal document. Where a figure is not published, the gate counts as unanswered. On the next page we apply the test to seven programmes in the pipeline or under way, using only the public record.

For investors the test reverses the usual order of due diligence. It looks at the crop and the anchor tenant before the land and the incentives, which usually lead the prospectus.

THE TEST APPLIED

Applied to seven programmes, the test is answered most often on the crop and least often on the anchor

EXHIBIT 22

What the public record shows against each gate, for seven programmes under way or planned

Heirstone Agro-Zone Test applied to information published by 2026

● evidence meets the gate ◖ partly met or mixed ○ evidence falls short – not published

PROGRAMME	1 CROP	2 POLICY	3 SIZE	4 ANCHOR	5 SUPPLY CHAIN	6 TARGETS
Agropole Nord, Senegal	● 80% of national rice and 90% of industrial tomato grown in the zone	– Not published	◖ Three parks of 130-150 ha; phasing not published	◖ Private sponsors provide 27% of financing	● 14 aggregation and service centres in the plan	○ 129,800 direct jobs, c.310 per park ha
Agropole Sud, Senegal	◖ Mango, cashew, maize; volumes not published	– Not published	◖ 25 industrial sites, 2025-2029	– Not published	– Not published	○ 350,000 jobs with Centre; XOF109bn announced, up from XOF57bn
SAPZ phase 1, Nigeria	– Hub crops not published in the sources read	– Not published	– Hub areas not published	– Not published	◖ Hubs with aggregation centres; Kaduna budget x53	○ 400,000 direct jobs (7-9 Mt cashew-equivalent); naira −73% over 40 months
SAPZ, Liberia	◖ Nine value chains named; volumes not published	– Not published	– Not published	◖ US$150m private investment targeted, no anchor named	◖ Transformation centres in Nimba, Bong and Grand Cape Mount	○ 156,000 jobs on US$19.45m, US$125 per job
Maranje park, Tanzania	● 528,263 t crop in 2024/25, c.2% processed	◖ US$0.15/kg export tax; minimum price; TMX auctions	○ 600,000 t capacity target, above the record crop	– Not published	◖ 406,362 t sold through TMX warehouse auctions	– Not published
Cashew cluster, Glo-Djigbé, Benin	◖ Crop over 190,000 t; c.70% diverted in 2024/25 (consultant estimate)	○ Raw exports banned from Apr 2024; export earnings −58% that year	● 15,000 t plant on 1.7 ha	● Arise IIP developer; five cashew units	○ 130,800 t capacity, 17,700 t processed (2023)	○ 300,000 jobs by 2030 in all sectors, 183 per ha
Kara agropole, Togo	◖ Rice, maize, soy; yields up 27-122%	◖ Target to lift processing share from 19% to 40%	○ 12,350 ha planned; 35% developed	◖ Four private operators; three firms selected for 3,000 ha	● Seven processing centres complete; 58,000 people with new water	– Not published

Source: Sources as in Exhibits 7, 13, 14 and 16; AfDB via AllAfrica (Nov 2023, Apr 2025); Le Soleil (Oct 2025); Ndarinfo (2025); AfDB Liberia appraisal (Sep 2022); TanzaniaInvest; Ecofin (Sep 2025); Benin Web TV (2022, 2026); GDIZ; AllAfrica FR (Aug 2018, Aug 2026); Agence Ecofin (2023). **Heirstone analysis.** An unanswered gate means the information was not in the public sources we found.

The crop gate is the one most programmes can answer, because the zones sit in producing regions: northern Senegal grows most of the country's rice and industrial tomatoes, and Tanzania produced over half a million tonnes of cashew in 2024/25. The anchor gate is the least often answered in public. Only Agropole Nord, with private sponsors in its financing, and Glo-Djigbé, with a private developer, show a committed private party from the start.

The targets gate is where programmes fall short on their own numbers. Five of the seven publish job targets above 150 per hectare or implying millions of tonnes of throughput. The zones may still succeed, but they will be judged against numbers they are unlikely to reach, which weakens the case for the next programme.

2 of 7

programmes with a private sponsor or developer committed in the public record

5 of 7

with job targets above observed densities or throughput

WHAT TO DO

Recommendations by reader

The evidence points the same way for every reader. Start from the crop and size the first phase to it, then report tenants and tonnes processed alongside hectares and jobs.

FOR

Governments and investment promotion agencies

North Africa and Sub-Saharan Africa

1. Size the first phase at 20 to 50 hectares and release the next only when 70% to 80% is sold or let. Souss Massa's 22 and 11 hectare tranches, now 90% sold, are the model.
2. Publish a yearly return for every zone: operating tenants, tonnes processed, capacity use and jobs on the payroll. Morocco's MEDZ already publishes a commercialisation rate.
3. Prefer graduated instruments to bans: an export tax, a factory price below the port price and allocation of raw material. If a ban is chosen, check first that installed capacity is being used.
4. Put aggregation, storage and cold chain in the same budget as the sheds. Sub-Saharan Africa loses 23% of food between harvest and retail.

FOR

Development finance institutions

AfDB, IsDB, IFAD, BOAD, GCF and co-financiers

1. Release park finance against tenants and tonnes processed as well as civil works, and report outcomes for each of the 27 SAPZ sites.
2. Appraise farmland programmes (US$10,000 to US$90,000 a hectare) separately from processing parks (US$0.2m to US$1.1m a hectare), with different results indicators.
3. Test job targets against observed benchmarks: 21 to 64 jobs per hectare in Moroccan agropoles; 44 to 56 per 1,000 t of cashew processed in Côte d'Ivoire.
4. Protect counterpart funding from devaluation by front-loading or indexing it. A naira budget lost 73% of its dollar value between approval and groundbreaking.

FOR

Agribusiness and food processors

Cocoa, cashew, grains, horticulture, dairy, oils

1. Locate where the crop and the price edge are. Côte d'Ivoire's floor-price ladder sets the factory floor FCFA30 per kg below the port floor.
2. Secure raw material and seasonal working capital before floor space. Processors at Yirgalem ran at 35% to 70% of capacity.
3. Treat land incentives as secondary. Land at US$1.2 per m² a year did not fill Ethiopia's parks; power, aggregation and crop finance set utilisation.
4. Model margins per tonne of raw material. Vietnam's kernels fetch 5.2 times the raw-nut price per tonne, but its kernel exports were worth 6% more than its nut imports.

FOR

Zone developers and investors

Private developers, PPP sponsors, lenders

1. Underwrite operating tenants per serviced hectare. Working agropoles reach 0.7 to 1.0 firms per hectare; Ethiopia's pilots had under 0.04 operating.
2. Price the policy behind the zone. Nkok's 144 investors arrived under a log-export ban and Glo-Djigbé's cashew units under a raw-nut ban: the mandate supports the zone and is also its main risk.
3. Sell or lease small plots. Loukkos sells from 2,000 m²; a 15,000 t cashew plant fits on 1.7 hectares.
4. Fund later phases from plot sales. Meknès drew MAD1.3bn of private investment on 35 hectares sold against a site cost of a little over MAD300m.

How Heirstone works on agro-industrial zones

We help governments, development finance institutions, agribusinesses and zone developers decide what to build and where: catchment and crop-volume studies, demand and tenant studies, highest-and-best-use and site selection, phasing strategy, feasibility and business models (DCF, IRR, NPV and Monte Carlo testing), masterplanning and the kind of benchmarking in this report.

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NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Agro-industrial zone: any designated area for processing agricultural products, including agropoles, integrated agro-industrial parks (IAIPs), special agro-industrial processing zones (SAPZs) and special economic zones with an agro-processing cluster. **Agropole:** in Morocco, a serviced estate for agri-food firms; in West Africa, often a territorial programme combining farmland, roads and a processing node. **RTC:** rural transformation centre, an aggregation point feeding an Ethiopian park. **RCN:** raw cashew nuts. **Processing share:** tonnes processed locally divided by production or marketed output, as defined by each source. **Operating tenant:** a firm reported as producing, as distinct from firms allotted land or installing.

Method

The Heirstone Agro-Zone Register compiles public figures for 20 zones and programmes to 2026. Local currencies are converted at the World Bank's annual average official rate (PA.NUS.FCRF) for the year stated, with euro amounts converted through the CFA franc peg (XOF655.957 per EUR); several currencies in the register moved sharply against the dollar between 2021 and 2025, so dollar figures from different years are not comparable without adjustment. Unit values are export value divided by quantity from UN Comtrade via WITS, per tonne of product. Per-hectare, per-job and per-tonne ratios are Heirstone analysis from the published inputs shown with them; where sources give different areas or budgets we show the range. Jobs per 1,000 t of cashew (43.6) divide 15,000 direct jobs by 344,026 t processed in Côte d'Ivoire in 2024; on 2023 tonnage (265,863 t) the ratio is 56. Map locations are approximate town centres; Bukanga-Lonzo and the New Delta are rough approximations. Single-source press figures are attributed in the text. Asking or target figures are not outcomes.

Use of this report

This report is published for general information. It is not investment, legal or valuation advice and should not be relied on as such. It uses public information only. © Heirstone Consulting L.L.C-FZ, 2026. Images are illustrative.

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4 PUSHING THE CROP INTO THE FACTORY

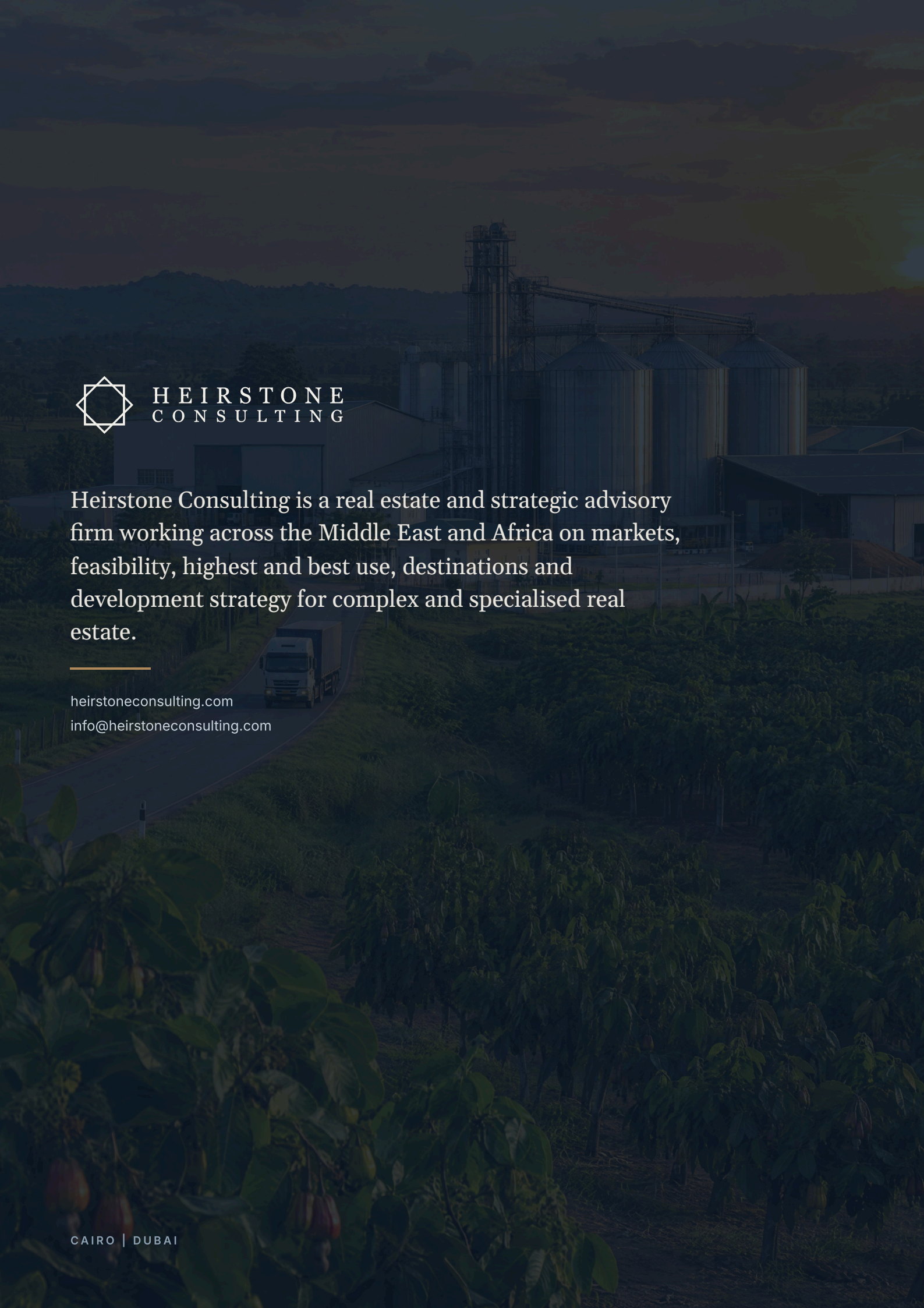
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